

# Ojas Makarand Deodhar

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## Education

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| <b>Arizona State University, Tempe, AZ, USA</b>                     | Aug 2024 – May 2026 |
| Master of Science in Computer Engineering (Computer Systems)        | GPA: 3.81/4.00      |
| <b>Vishwakarma Institute of Information Technology, Pune, India</b> | Aug 2019 – May 2023 |
| Bachelor of Technology in Computer Engineering                      | GPA: 3.86/4.00      |

## Technical Skills

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| <b>Programming Languages:</b> Python, Bash, Java, C/C++, SQL                                                                   |
| <b>Tools &amp; Platforms:</b> AWS, Docker, Git, FastAPI, Flask, Streamlit, LangChain, Power BI, Microsoft Excel, Chroma, FAISS |
| <b>AI/ML:</b> TensorFlow, PyTorch, Keras, Scikit-Learn, HuggingFace, Transformers, Habana (Intel Gaudi), CUDA, RAG, LLM        |
| <b>Python Libraries:</b> NLTK, NumPy, Pandas, SciPy, BeautifulSoup, OpenCV                                                     |

## Experience

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| <b>Data Analyst (SW IV), Arizona State University – Decision Theater Network, Tempe, AZ</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | May 2025 – May 2026 |
| <ul style="list-style-type: none"><li>Built an LLM-powered analytics pipeline using TableGPT-2, LangChain, and Pandas that let users query census, geospatial, and health datasets with natural language, automatically extract data with Python, and validate outputs, resulting in quicker insight generation for research projects</li><li>Optimized LLM inference pipelines on Intel Gaudi 2 HPUs and NVIDIA GPUs within ASU's SOL HPC environment using optimum-habana, improving throughput and scalability for large geospatial workloads</li><li>Built a geospatial health-facility finder using the OpenRouteService API, Haversine distance calculations, and GeoDataFrame filtering, and displayed results on interactive Streamlit and Folium dashboards with shapefile overlays, allowing users to quickly locate and compare the nearest hospitals, urgent-care centers, and clinics</li><li>Integrated Gemini and OpenAI APIs to translate user-reported symptoms into semantically matched nearby healthcare facilities, delivering triage recommendations that guide users to appropriate care locations</li><li>Engineered a RAG pipeline using Gemini, ChromaDB, and cosine similarity to score course-skill relevance and automatically generate custom syllabi from job-market skill clusters, helping educators create more targeted curricula aligned with industry needs</li></ul> |                     |

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| <b>Machine Learning Engineer Jr., DigitalMain Pvt. Ltd., Pune, India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | June 2023 – June 2024 |
| <ul style="list-style-type: none"><li>Trained classification model, aided with BERT embeddings as features to classify interrogative text, improving Slack thread classifier and automated Q&amp;A generation with accuracy of 92%</li><li>Optimized search re-ranking by training and evaluating BERT Cross-Encoder models (BGE, Cohere, BERT model variants (like DistilBERT, RoBERTa), custom trained), with BGE Reranker Large achieving the highest accuracy of 93%</li><li>Developed Python-based data preprocessing and inference pipelines using Pandas and Scikit-learn, which streamlined ML workflows and reduced data-preparation time, enabling scalable model deployment</li><li>Evaluated 5+ Large Language Model APIs (ChatGPT 3.5, LLaMA 2, Gemini, Cohere, Claude, etc.) to improve summarization, Q&amp;A, and RAG-based systems, resulting in more accurate and coherent generated content across projects</li><li>Designed LLM prompts for Retrieval Augmented Generation, text summarization, Slack thread classification, and automation of 5+ RFP and SoW documents using Python and the OpenAI API and built Streamlit proof-of-concepts that demonstrated feasibility to stakeholders, accelerating adoption</li></ul> |                       |

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| <b>Machine Learning Intern, DigitalMain Pvt. Ltd., Pune, India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Aug 2022 – May 2023 |
| <ul style="list-style-type: none"><li>Built data pipelines with Pandas and PyPDF2 to extract and preprocess text from PDFs and web pages, streamlining data preparation for BERT-based model training</li><li>Evaluated summarization models (BERT, T5, BART, GPT-2) using BLEU and BERTScore with the Transformers library, leading to the selection of the best model for the project</li><li>Prepared and tagged Slack datasets for embedding and classification models, improving downstream performance and production pipelines</li></ul> |                     |

## Projects

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| <b>Automated Resume Screening Tool</b>                                                                                                                                                                                                                                                               | <a href="#">GitHub Repo</a>       |
| <ul style="list-style-type: none"><li>Built a Streamlit app using Mistral APIs + LangChain to evaluate resumes vs job descriptions with automated scoring and brief feedback. Multiple resumes can be uploaded for a job role.</li></ul>                                                             |                                   |
| <b>Loan Default Prediction — LendingClub Data</b>                                                                                                                                                                                                                                                    | <a href="#">GitHub Repo</a>       |
| <ul style="list-style-type: none"><li>Trained ML models (RF, XGBoost, LightGBM) achieving 74% accuracy; evaluated with AUC, F1, precision/recall. Along with this, the project involved handling data issues like missingness, class imbalance, feature selection, and feature engineering</li></ul> |                                   |
| <b>AI_Blur_Studio — Real-time Background Effects</b>                                                                                                                                                                                                                                                 | <a href="#">HuggingFace Space</a> |
| <ul style="list-style-type: none"><li>Developed a Gradio app deployed on Hugging Face Space for photo segmentation with a human, using a Segmentation model, applying a Zoom-style video conferencing effect, and a realistic camera depth of field using a depth estimation model</li></ul>         |                                   |

## Publication

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| <ul style="list-style-type: none"><li><i>Shark Tank Deal Prediction using Machine Learning Techniques</i> DOI: 10.1109/I2CT61223.2024.10543650<br/>Ojas Deodhar, Sarthak Bhatkar, Pradnya Dixit, Preeti Kulkarni, Atharv Oak, Shreenivas Londhe<br/>IEEE 9th International Conference for Convergence in Technology, April 2024</li></ul> |  |
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